From Detection to Mitigation: Addressing Bias in Deep Learning Models for Chest X-Ray Diagnosis

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Introduction

Deep learning (DL) models have demonstrated significant potential in improving chest X-ray (CXR) diagnosis. However, these technologies risk exacerbating healthcare disparities if their performance varies across different subgroups of patients, for example according to sex, age, and race. Addressing the inherent biases of DL models is essential to ensure their safe and reliable deployment in clinical practice. Current bias mitigation methods require extensive model retraining and are thus impractical in healthcare due to data scarcity and resource constraints.

We show that replacing the final layer of Convolutional Neural Network (CNN) with an eXtreme Gradient Boosting (XGBoost) classifier improves the fairness across subgroups while improving the overall predictive performance.

We find that our approach offers competitive bias reduction in comparison to traditional full-model training bias mitigation techniques at a fraction of the computational cost.

We find that combining XGBoost retraining with active learning yields the largest reduction in bias across all demographic subgroups.

Data and Pretrained Model

CheXpert: 224,316 CXRs obtained at Stanford Health Care
MIMIC-CXR: 377,110 CXRs from the Beth Israel Deaconess Medical Center

CNNs: DenseNet-121 and ResNet-50 models from TorchXrayVision, trained on the CheXpert dataset to predict 18 chest medical conditions

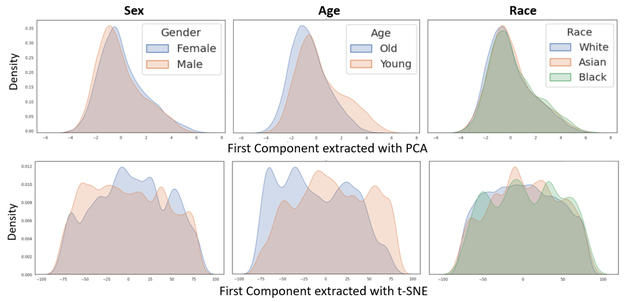
Metrics to Assess Bias

1. Having a good overall performance: AUPRC provide a good trade-off between precision and recall and is effective when dealing with imbalanced datasets (higher is better).
2. Having similar performance across subgroups: Δ AUPRC (lower is better).

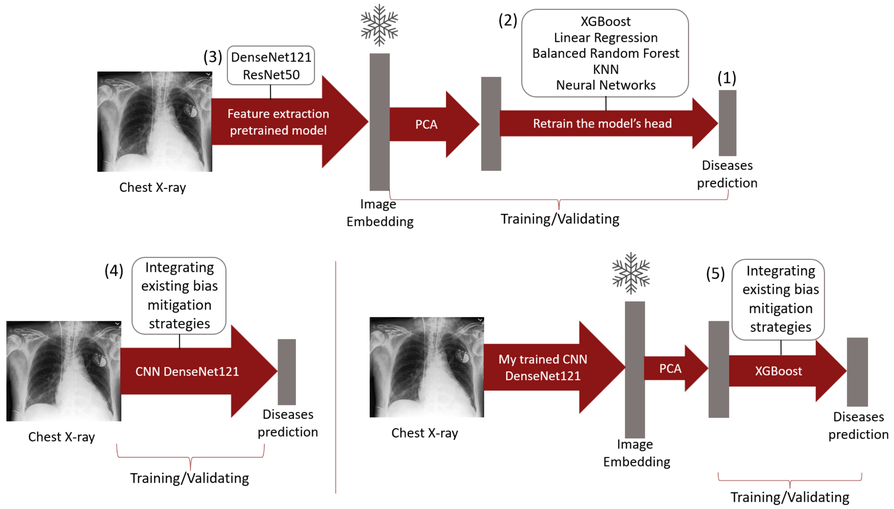
Bias Detection

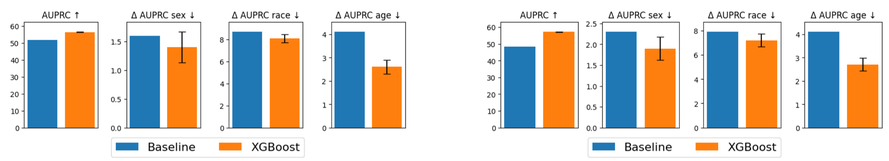
To better understand the origins of bias, we analyzed three potential sources:

- **Data and Labels:** Strong imbalances across race and age may introduce confounding if unaddressed.
- **Model Representations:** PCA/t-SNE of embeddings showed separation by sex, age, and race. A classifier trained on embeddings achieved high accuracy, confirming encoded demographic signals.
- **Radiologist Variability:** Analysis of 500 cases across eight radiologists showed no subgroup-specific uncertainty, suggesting label noise is not a major bias source.

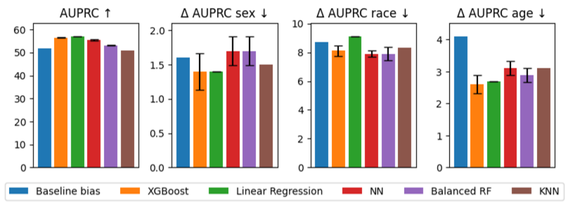


Bias Mitigation

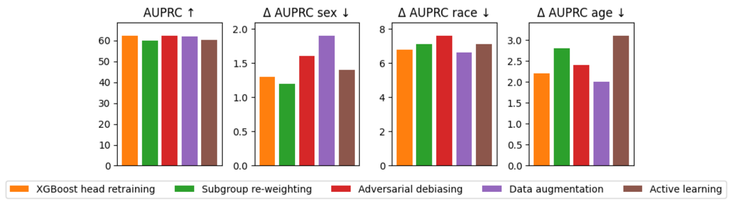




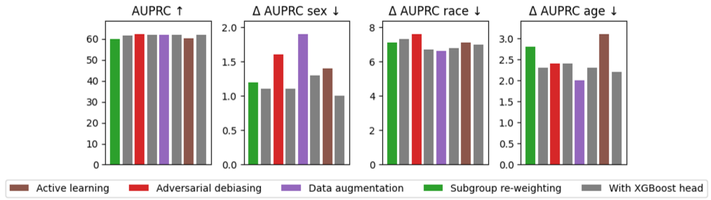
1-3) Performance and bias mitigation of our XGBoost head retraining method for multiple medical conditions classification applied to different model architectures to extract embeddings, DenseNet on the left, ResNet on the right.



2) Performance and Bias mitigation of different ML models to retrain the head.

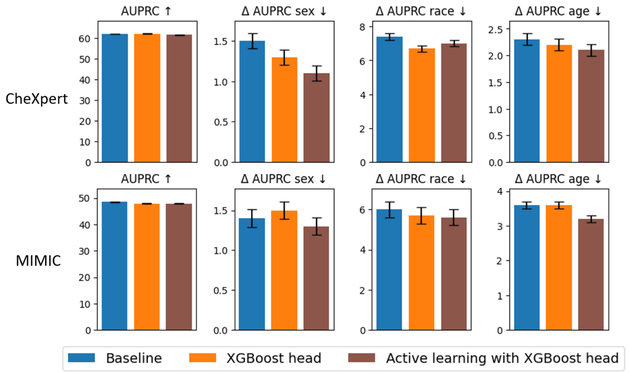


4) Comparison of our new lightweight bias mitigation method (in orange) with existing methods that require full model retraining.



5) Comparison of existing bias mitigation methods that require full model retraining with combine the method with our XGBoost head retraining (in grey).

Clinical Significance



Comparison of initial performance and bias (baseline in blue) with XGBoost head retraining (orange), and XGBoost head retraining combined with active learning (brown), ID on CheXpert and OOD on MIMIC.

To minimize False Negative Rates (FNR), thresholds were chosen before and after bias mitigation to ensure recall > 0.95.

	ΔFNR	ΔTPR	ΔFPR	EO max gap
Before mitigation	0.023	0.023	0.015	0.023
After mitigation	0.010	0.010	0.007	0.010

Performance metrics before and after bias mitigation for Pleural Effusion detection in women across racial subgroups.

Bias mitigation reduced disparities in both sensitivity and specificity by about half, translating into fewer missed cases and fairer outcomes. Clinically, this means patients receive more consistent diagnostic performance across racial groups, increasing both reliability and trust in the model. Such improvements also enhance the likelihood of clinician adoption of AI tools that demonstrate equitable behavior across diverse populations.

Conclusion and Future Work

In this work, we introduced a practical and lightweight framework for detecting and mitigating demographic bias in DL models for CXR diagnosis. By replacing the final classification layer of a CNN with an XGBoost model, we demonstrated that it is possible to significantly reduce disparities across sex, age, and race subgroups while preserving, if not improving, overall model performance. Our approach demonstrated robustness across multiple medical conditions and datasets, maintaining performance both ID (CheXpert) and OOD (MIMIC). Additionally, our method outperformed traditional bias mitigation techniques, including data augmentation, active learning, and adversarial training. Combining our XGBoost head retraining with active learning resulted in an optimal balance between fairness and diagnostic accuracy.

Future work will extend this framework to other architectures, imaging modalities, and tasks beyond classification, enabling broader application of fairness-aware AI in healthcare.

Code availability

Our code is publicly available on our GitHub. The repository includes documentation and scripts to adapt our bias detection and mitigation framework to new datasets and tasks.

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